

Peer Review Process, Evaluation Criteria and Rating Classifications

All eligible grant applications are reviewed by appropriate external technical peer reviewers based on the criteria and process described below. This review is designed to evaluate each application according to its scientific merit. The individual external peer reviewers include non-EPA scientists, engineers, social scientists, and/or economists who are accomplished in their respective disciplines and proficient in the technical subjects they are reviewing.

Prior to the external technical peer review panel meeting, all reviewers will receive electronic copies of all applications, as well as a full set of abstracts for the applications. Each application will be assigned to a minimum of three primary peer reviewers, one of whom will be assigned the role of Rapporteur. Each reviewer will be assigned up to approximately 10 applications on which to serve as a primary reviewer. During the review period leading up to the panel meeting, primary reviewers will read the full set of abstracts and entire application package for each application they are assigned. They will also prepare a written individual evaluation for each assigned application that addresses the peer review criteria described below and rate the application with a score of excellent, very good, good, fair, or poor.

At the beginning of the panel meeting, each primary reviewer will report their ratings for the applications they reviewed. Those applications receiving at least two ratings of *Very Good* or one rating of *Excellent* from among the primary reviewers will then be further discussed by the panel in terms of the peer review criteria below. In addition, if there is one *Very Good* rating among the primary reviewers of an application, the primary reviewer, whose initial rating is the *Very Good*, may request discussion of the application by the peer review panel. All other applications will be declined for further consideration.

After the discussion of an application by the panel, the primary reviewers may revise their initial ratings and if they do so, this will also be documented. The final ratings of the primary reviewers will then be translated by EPA into the final peer review score (excellent, very good, good, fair, or poor) for the application. This is reflected in a peer review results document developed by the Rapporteur which combines the individual initial and final evaluations of the primary reviewers and captures any substantive comments from the panel discussion. This score will be used to determine which applications undergo the internal programmatic review discussed below. A peer review results document is also developed for applications that are not discussed. However, this document is a consolidation of the individual primary reviewer initial evaluations, with an average of the scores assigned by the primary reviewers.

Peer reviewers consider an application's merit based on the extent to which the application demonstrates the criteria below. Criteria are listed in descending order of importance (i.e., Criteria 1 has the heaviest weight).

1. Research Merits: (subcriteria are in descending order of importance):

- a. The degree to which the application demonstrates that the research is original and contributes to the scientific knowledge in the topic area. And the degree to which the application demonstrates that the project (and its approach) is defensible and technically feasible, and uses appropriate and adequate research methods.
- b. The degree to which the application demonstrates that the project results will produce benefits to the public (such as improvements to the environment or human health) and will be disseminated to enhance scientific and technological understanding.

2. Responsiveness: The degree to which the application demonstrates that the research is responsive to the objectives, research needs, and special considerations specified by the RFA.

3. Project Management (subcriteria are equally weighted):

- a. Investigators: The degree to which the application demonstrates that the Principal Investigator(s) and other key personnel have the appropriate qualifications (including research training, demonstrated knowledge of pertinent literature, experience, and publication records).
- b. Management: The degree to which the application demonstrates that the project will be adequately managed to ensure the timely and successful achievement of objectives using appropriate project schedules and milestones. And the degree to which the application demonstrates the applicant will adequately track and measure progress toward achieving expected results (outputs and outcomes).
- c. Quality Assurance (QA): The degree to which the application includes an appropriate and adequate QA Statement.
- d. Resources and Cost Controls: The degree to which the application demonstrates that the facilities, equipment, and budget are appropriate, adequate, and available. And the degree to which the application demonstrates that well-defined and acceptable approaches, procedures, and controls are used to ensure timely and efficient expenditure of awarded grant funds.

4. Other Factors (subcriteria are equally weighted):

- a. Innovation: The degree to which the application demonstrates that the research will challenge and seek to shift current research or engineering paradigms by using innovative theoretical concepts, approaches or methodologies, instrumentation or interventions applicable to one or more fields of research.
- b. Sustainability: The degree to which the application demonstrates that the research will embody the principles of sustainability and seek sustainable solutions that protect the environment and strengthen our communities. The sustainability primer (see link) provides examples of research activities that promote and incorporate sustainability principles ([Sustainability Primer \(PDF\)](#) (2 pp, 195 K)).
- c. Community Engagement and Communications Plan (CECP): The degree to which the CECP clearly describes how communities and stakeholders will be involved in the research and implementation processes, and describes planned interactions with these partners through the course of the project. The degree to which the CECP effectively ensures that communities are supportive of the proposed research and empowered to take action to reduce pollutant exposure, either immediately or in the future. The degree to which the methods of communication and the levels of community engagement are appropriate for the research topic and are likely to have a tangible benefit for communities. The degree to which the applicant has described: [1] an active partnership with a community or community-based organization, [2] the partner's intent to participate in the proposed research and [3] that the applicant or partner has acquired enough data to identify and prioritize public-health and ecological concerns. Applicants that do not plan on partnering with other groups in the performance of the project will be evaluated based on the extent to which they demonstrate how they will be able to effectively perform and complete the project without such partnership.

Ratings Classifications

Excellent: A truly outstanding proposal with numerous exceptional attributes.

Very Good: A very strong proposal with many noteworthy merits.

Good: A solid proposal but not among the best, may have some deficiencies.

Fair: Not a strong proposal.

Poor: A proposal with serious deficiencies.